

TELECOM CHURN ANALYSIS

26.54% Churn Rate	Month-to-Month Highest Risk	1–2 Months Critical Window	E-Check Users Top Payment Risk
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Exploratory Data Analysis | Summary Report

WHAT WAS ANALYSED

- Dataset loaded; shape, info, describe checked
- TotalCharges cleaned (blanks → 0, changed datatype to float)
- Null check & duplicate customerID check — data clean
- SeniorCitizen column encoded: 0/1 → Yes/No
- Overall churn: countplot + pie chart → 26.54% churn
- Churn by gender — no significant difference found
- Senior citizen churn — higher churn rate confirmed
- Tenure histogram — early customers churn heavily
- Contract type — month-to-month shows the highest churn
- 9-service grid & payment method analysis done

DEFECTS IN PLANNING & EXECUTION

!	Short-term Contract Strategy	Over-reliance on month-to-month contracts — the highest churn tier. No effective incentive structure to retain customers in longer plans.
!	No Early Onboarding Process	Churn is concentrated in months 1–2. Customers aren't anchored early enough — no structured welcome, value demo, or check-in process exists.
~	Poor Sticky Service Adoption	Customers without OnlineSecurity, TechSupport, OnlineBackup, and DeviceProtection churn more. These features are not proactively cross-sold or bundled.
~	E-Check Users Unaddressed	Electronic Check users churn the most. This high-risk segment has no targeted outreach, flexible payment option, or migration incentive from the company.
~	Senior Citizens Underserved	Seniors churn at a higher rate than non-seniors. No dedicated support structure, simplified plan, or accessibility feature exists for this demographic.

MEASURES TO REDUCE CHURN

1	Incentivise Long-term Contracts	Offer 10–20% discounts on annual/2-year plans. Bundle exclusive perks (extra data, add-ons) only for longer commitments. Prompt upgrades at the 3-month mark.
2	30-Day Onboarding Programme	Day 1 welcome call, Day 7 usage check-in, Day 14 feature spotlight (security, backup), Day 30 satisfaction survey. Guided onboarding cuts early churn significantly.
3	Bundle Sticky Services by Default	Include OnlineSecurity and TechSupport in mid/premium plans by default. Offer 3-month free trials for basic-plan users. More services = lower churn propensity.
4	Target E-Check Users with Offers	Offer a bill credit or data bonus to Electronic Check users who switch to auto-pay. Flag them as a priority retention segment and reduce payment friction.
5	Senior Citizen Dedicated Plan	Design a simplified Senior Plan with dedicated phone support, easy billing, and loyalty discounts. Proactive monthly check-in calls for senior accounts.
6	Build a Churn Prediction Model	Train Logistic Regression or Random Forest on this dataset to score monthly churn risk. Flag high-risk customers for proactive retention offers before they leave.

Analysis performed using Python (Pandas, Matplotlib, Seaborn) | Dataset: Customer Churn.csv (Kaggle)